

Response to the Editor and Reviewers

Original manuscript ID: IoT-65990-2026

MOS²: A Two-Stage Multi-Objective Framework for Server Deployment and Service Provisioning in Mobile Edge Computing

Revision format. All changes are highlighted in blue in the marked manuscript; a clean manuscript is provided separately.

Dear Dr. Zhang and Reviewers,

Thank you for the careful evaluation and constructive comments. We have revised the manuscript comprehensively and provide point-by-point responses below. Reviewer comments are reproduced verbatim before each response. Where the manuscript was changed, the exact revised passage is reproduced in a highlighted box; additional experimental claims are accompanied by the corresponding figures or numerical evidence.

Principal revisions

- redesigned Fig. 1 and clarified every region, link, and service symbol;
- renamed all deployment, provisioning, and association variables consistently;
- justified the two-stage formulation and quantified a small-scale exact joint-optimization gap;
- made the PSP capacity-repair mechanism explicit in Fig. 2, Algorithm 3, and the text;
- re-exported experimental figures with readable labels, ticks, and consistent panel dimensions;
- added CLS initialization sensitivity, complete PSP parameters, Q normalization, and Pareto metrics;
- included a DQN learning-based provisioning baseline; and
- added a real-region two-stage generalization experiment as response evidence.

Response to Associate Editor

Associate Editor Comment

Associate Editor Comment

This is an extended version of a conference paper. I can see there are quite some new contents for algorithm, experiments, etc., so a submission to a journal is reasonable to me. We got the comments from two reviewers, both offered detailed comments and overall recommendations, that are not positive enough for an acceptance. I recommend a reject, but meanwhile suggest to offer the authors a chance to revise significantly and re-submit; however the authors can also consider submitting this work to other more suitable journals. Note that the authors shall feel free to evaluate each reference and only cite those with true and big enough relevance to this study.

Response

Thank you for recognizing the journal-level extension and for allowing a substantially revised resubmission. We have addressed every technical and presentation issue raised by the two reviewers. The marked manuscript highlights all changes in blue, and a clean version is provided separately. The revision includes a redesigned system illustration and algorithm flow, a unified notation system, a technical rationale for the two-stage formulation, an explicit feasibility-repair procedure, additional reproducibility details, a CLS initialization-sensitivity study, a DQN baseline, quantitative Pareto metrics, and focused supporting experiments.

We also followed the guidance on references. Each of the six suggested works was evaluated individually. Four directly relevant studies on service deployment/offloading, local search, long-term MEC resource allocation, and online learning were incorporated into the Related Work section. The two FSO-enabled SAGIN task-offloading studies were not cited because their network architecture, link model, and decision variables differ substantially from the terrestrial capacity-constrained service-provisioning problem considered here.

Response to Reviewer 1

Comment 1: Clarity of Fig. 1 and the dashed regions

Reviewer Comment

Figure 1 in the paper is somewhat cluttered and difficult to follow. Specifically, the regions enclosed by the dashed lines are not clearly explained. The authors should refine the visual layout of this figure and provide a more explicit description in both the text and the caption regarding what these dashed areas represent.

Response

Thank you for identifying the ambiguity in Fig. 1. We revised the illustration and its accompanying explanation so that every visual element has a unique meaning. The solid black contours partition the topology into interconnected service regions; the black dashed circles denote individual base-station coverage areas; the red dotted circles identify base stations equipped with edge servers; blue bidirectional arrows represent cloud-edge communication; and red dashed arrows indicate inter-region request forwarding. User colors correspond to requested service types, and the service blocks beside an edge server identify its instantiated services. The question-mark slot in Area 3 deliberately represents the remaining provisioning decision under limited capacity.

The caption is now a concise one-sentence description, while the main text explains the dashed regions, routing arrows, service colors, instantiated services, and the open service slot in detail.

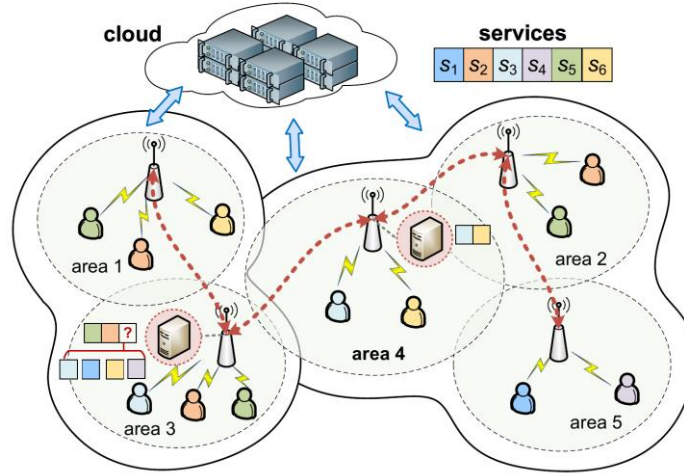
Revised Manuscript Text

Location: Section I-A, Fig. 1 and the accompanying discussion

Fig. 1. MEC system model and service-provisioning scenario.

Server deployment and service provisioning have a significant impact on the profit of mobile network operators. Figure 1 illustrates a scenario with five base stations. The solid black contours partition the topology into interconnected service regions, the black dashed circles denote the coverage areas of individual base stations, and the red dotted circles identify base stations equipped with edge servers. The blue bidirectional arrows represent cloud-edge communication, while the red dashed arrows indicate inter-region request forwarding. Distinct colors represent service types $S_1 - S_6$, and each user color indicates the corresponding requested service. The service blocks beside an edge server show its instantiated services; the question-mark slot in Area 3 denotes a provisioning decision under limited server capacity.

Each service replica serves a user request and generates income. Because an edge server can host only a subset of the service catalog, a request without a local replica is forwarded to another server that provides the requested service, incurring additional transmission cost. For example, the server in Area 4 provisions services S_3 and S_6 ; requests for the other service types require inter-area forwarding.



Response Fig. R1-1. Revised system model and service-provisioning scenario.

Comment 2: Conflicting decision-variable notation

Reviewer Comment

There are some conflicts and visual similarities in the defined notations, particularly concerning the decision variables. For example, x_{ijw} and x_{jw} are used to denote different concepts, which may easily confuse the readers. It is strongly recommended to use completely distinct variable letters to differentiate the decision variables clearly.

Response

We agree that the original notation could cause confusion. The decision variables have been renamed consistently throughout the system model, Table I, constraints, objective functions, and algorithm descriptions: y_{jk} denotes server deployment, z_{jw} denotes service provisioning, and a_{ijw} denotes user-service-server association. The implementation entry $\text{indiv}[j, w]$ is explicitly defined as the encoded counterpart of z_{jw} . This revision removes the former visual conflict among x_{jk} , x_{jw} , and x_{ijw} and also eliminates the inconsistent use of w_j for server m_j .

Revised Manuscript Text

Location: Section III-A and Section IV-B

Equation (7) defines the server-deployment variable y_{jk} , service-provisioning variable z_{jw} , and user-service association variable a_{ijw} .

A service-provisioning scheme is encoded by a binary matrix $\text{indiv} \in \{0,1\}^{k \times |\mathcal{S}|}$. Each row corresponds to a deployed server and each column to a service type. The entry $\text{indiv}[j, w]$ is the encoded counterpart of z_{jw} : it equals 1 when service S_w is provisioned on server m_j , and 0 otherwise.

Notation revision summary

Decision	Revised variable	Definition
Server deployment	y_{jk}	Server m_j is deployed at base station b_k
Service provisioning	z_{jw}	Service s_w is provisioned on server m_j
User association	a_{ijw}	User u_i is served by s_w on m_j

Comment 3: Rationale and potential optimality loss of the two-stage decomposition**Reviewer Comment**

The authors transform the highly intertwined server deployment and service provisioning problem into two relatively independent stages. While this decoupling effectively reduces computational complexity, it raises the question of whether this transformation leads to a loss of global optimality. The authors should add a discussion justifying this two-stage decomposition and, if possible, comment on or analyze the potential performance gap compared to a joint optimization approach.

Response

Thank you for raising this important point. The revised manuscript explains that the decomposition follows the operational hierarchy of MEC planning. Server deployment is a relatively long-term infrastructure decision governed by budget, coverage, and geographic demand, whereas service instances are provisioned after physical server locations are known. Stage I passes the selected locations, user assignments, and deployment-related cost to Stage II, thereby preserving the principal cost-latency dependence while substantially reducing the combinatorial decision space.

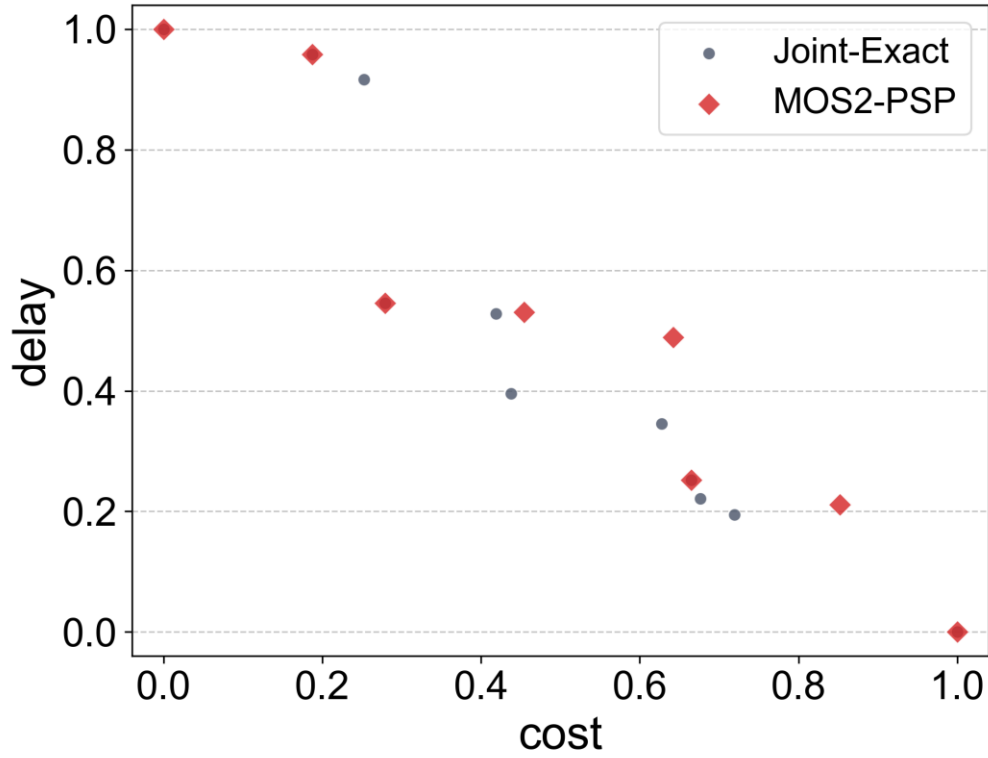
We further conducted a response-only exact small-scale joint-optimization experiment. The instance contains 6 candidate stations, 3 deployed servers, 30 users, 4 service types, and a per-server service capacity of 2. All $\binom{6}{3} \times \left[\sum_{r=0}^2 \binom{4}{r} \right]^3 = 26,620$ feasible joint server-service decisions were enumerated to construct an exact Pareto reference. Across seeds 42, 43, and 44, MOS²-PSP attained the same best normalized weighted quality as the exact joint reference in every run. The mean HV gap was 3.87%, the mean IGD was 0.0382, and exact enumeration required 5.18 times the MOS² runtime on average even at this reduced scale.

The near-identical balanced cost-delay quality and the steep growth of exhaustive search show that decomposition is not merely a computational convenience, but is necessary for tractable optimization at the larger MEC scales studied in the manuscript. MOS² preserves the decisive deployment-to-provisioning dependencies while matching the exact solver's best balanced solution in every tested seed, using only about one-fifth of its runtime and avoiding enumeration and storage of the full joint decision space.

Revised Manuscript Text

Location: Section III-D, decomposition rationale

The decomposition follows the operational hierarchy of MEC planning. Server deployment is a long-term infrastructure decision governed by budget, coverage, and geographic demand, whereas service instances are provisioned after the physical server locations are known. Stage 1 therefore resolves the spatial variables that determine deployment and transmission costs, and Stage 2 optimizes service instances and user associations on the selected infrastructure. Passing the selected locations, user assignments, and Stage-I cost to Stage 2 preserves the cost-latency dependence while substantially reducing the decision space of the service-provisioning search.



Response Fig. R1-2. Representative small-scale Pareto comparison between Joint-Exact and MOS²-PSP (seed 42; response-only evidence).

Small-scale exact joint-optimization results

Seed	HV gap	IGD	Best Q gap	Runtime Joint/MOS ²
42	6.45%	0.0289	0.0000	4.99x
43	4.18%	0.0199	0.0000	5.15x
44	0.98%	0.0660	0.0000	5.40x
Mean	3.87%	0.0382	0.0000	5.18x

Comment 4: Constraint checking and handling in PSP

Reviewer Comment

The PSP strategy was proposed for the service provisioning stage, the logic surrounding the constraint check step is somewhat vague and unclear. The authors need to adjust this part of the diagram and elaborate in the corresponding text on exactly how the constraints are verified and handled during the algorithm's execution (e.g., whether penalty functions or repair mechanisms are used).

Response

Fig. 2, Algorithm 3, and the associated text now describe the complete feasibility procedure. After crossover and mutation, each decision entry is rounded to a binary value. For every server m_j , PSP checks whether $\sum_w z_{jw} \leq V_j$. If capacity is exceeded, selected service entries are randomly deactivated until feasibility is restored. Objective evaluation and non-dominated sorting are performed only after repair. PSP therefore

enforces capacity through an explicit repair mechanism rather than a penalty function. The redesigned Evolutionary Optimization panel presents the same sequence and uses proper mathematical subscripts for the population and capacity variables.

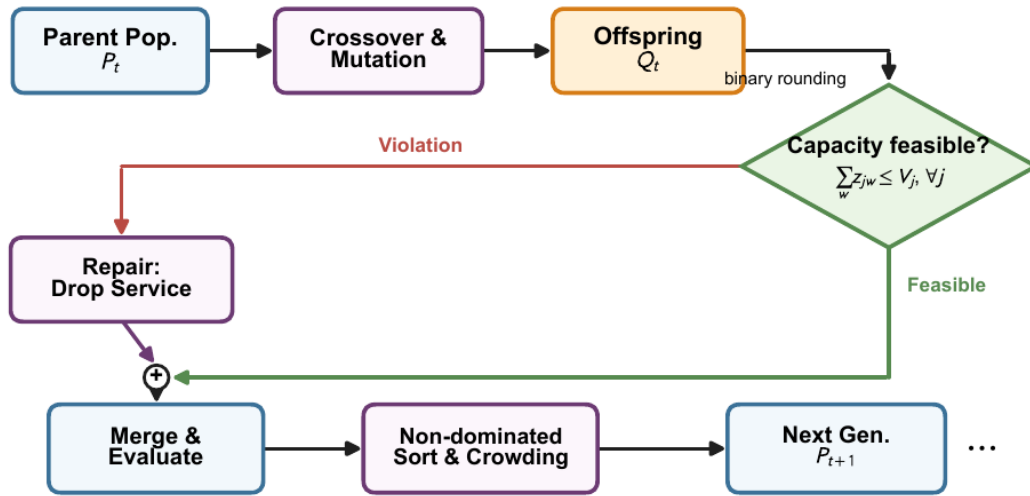
Revised Manuscript Text

Location: Section IV and Algorithm 3

Fig. 2. Overall architecture of MOS².

Given the fixed server locations, PSP constructs a hybrid initial population and applies NSGA-II to optimize provisioning cost and access delay. Crossover and mutation produce each offspring, whose entries are first rounded to binary values. For every server m_j , PSP then checks $\sum_w z_{jw} \leq V_j$. If the capacity is exceeded, selected service entries are randomly deactivated until feasibility is restored. Only capacity-feasible offspring are evaluated and passed to non-dominated sorting; thus, feasibility is enforced by repair rather than by a penalty function. The evolutionary process returns a set of provisioning schemes representing different cost-delay trade-offs.

After crossover and mutation, every encoded entry is rounded to 0 or 1. If $\sum_w \text{indiv}[j, w] > V_j$ for any server m_j , selected entries in that row are randomly set to 0 until the capacity constraint is satisfied. Objective evaluation and non-dominated sorting are then performed on the repaired feasible individual.



Response Fig. R1-3. Revised Evolutionary Optimization flow showing binary rounding, capacity checking, repair, evaluation, and population update.

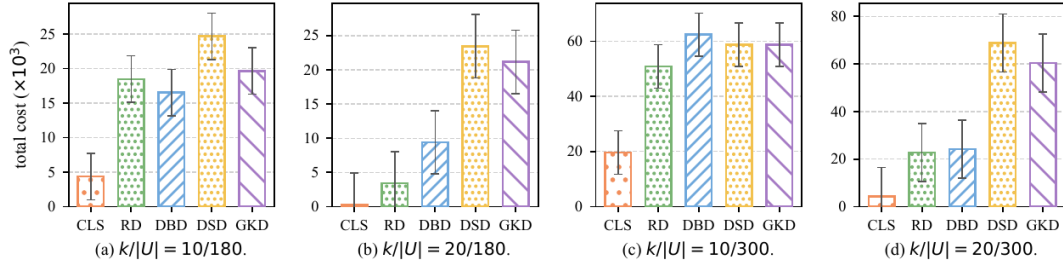
Comment 5: Readability of axes, labels, and ticks

Reviewer Comment

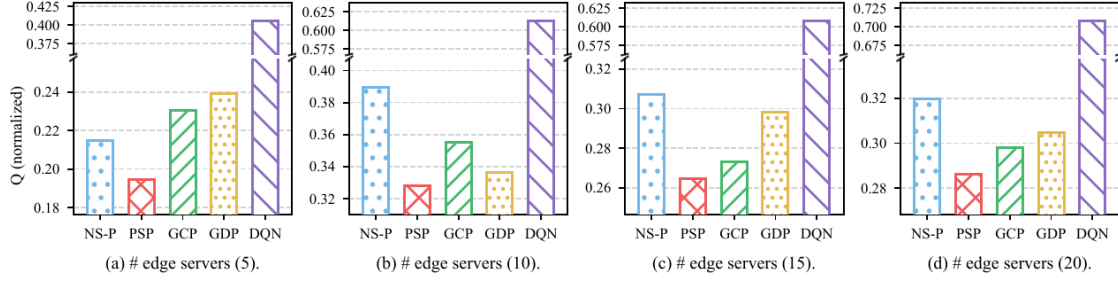
The font sizes for the X-axis and Y-axis labels/ticks in all experimental figures are too small. Please enlarge them to ensure they are easily readable.

Response

We re-exported the experimental figures with consistent multi-panel dimensions, larger axis labels and tick labels, improved panel spacing, and concise captions. The Stage-I scale and convergence results, Stage-II scalar comparisons, hybrid-initialization illustration, Pareto fronts, and CLS sensitivity figure were all visually checked at the final IEEE column widths. Redundant shared legends and auxiliary annotations were removed where they reduced usable plotting area.



Response Fig. R1-4. Re-exported Stage-I multi-panel result.



Response Fig. R1-5. Re-exported Stage-II multi-panel result at double-column width.

Comment 6: Typographical and language errors

Reviewer Comment

There are several typos in this paper. A thorough proofreading is required. Some specific examples include: page 5: s_2 , s_3 and s_5 . $\rightarrow s_2$, s_3 , and s_5 ; page 12: U the set of $\rightarrow U$ is the set of.

Response

The manuscript has been proofread throughout. The two examples identified by the reviewer were corrected. We also corrected ordinal suffixes in the author affiliations, subject-verb agreement in the Introduction, the phrase 'hierarchy network architecture,' repeated introductory wording, inconsistent server symbols, punctuation in service lists, and grammatical issues in the system-model and algorithm descriptions. The marked and clean manuscripts were both compiled and visually inspected after proofreading.

Revised Manuscript Text

Location: Sections I and III, representative corrected passages

In this paper, we consider a hierarchical network architecture comprising the cloud data center, edge servers, and mobile end users, as illustrated in Figure 1.

At the user layer, the set of mobile users is denoted by $\mathcal{U} = \{u_i\}$, where u_i represents the i -th user.

Representative proofreading corrections

Original	Revised
s2 , s3 and s5	s2, s3, and s5
U the set of	U is the set of
4rd / 5rd	4th / 5th
hierarchy network architecture	hierarchical network architecture

Response to Reviewer 2**Comment 1: Initialization sensitivity of CLS****Reviewer Comment**

The proposed CLS algorithm is essentially a local-search-based heuristic for the K-median problem. However, the manuscript does not clearly explain the initialization sensitivity of the algorithm. Since the initial deployment set S in Algorithm 1 is randomly generated, different initializations may lead to significantly different local optima.

Response

We added a 50-run initialization-sensitivity study comparing Random, Density, Distance-Sum, marginal Greedy, and Diverse initializations. For each server/user configuration, the reported gap is the percentage difference between the final CLS cost and the best final cost observed under the same configuration.

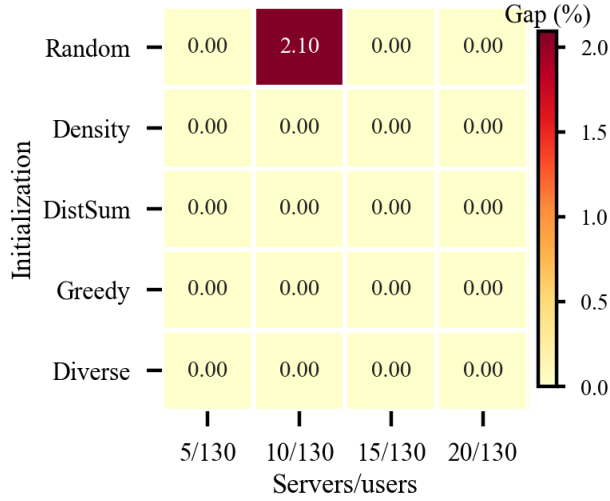
With 130 users and 5, 10, 15, or 20 deployed servers, Random reached the same best final cost as the specialized initialization strategies in three of the four configurations and showed only a small 2.10% mean gap in the remaining case. More importantly, in the complementary 10-server/150-user stress case, Random obtained a 1.27% mean gap, whereas marginal Greedy deteriorated to 15.88%. Thus, handcrafted initialization provides no consistent optimization advantage, and a deterministic greedy preference can even steer CLS toward a markedly poorer local optimum. We therefore retain Random initialization because it matches the dedicated strategies in most tested settings while remaining more robust to initialization bias.

Revised Manuscript Text

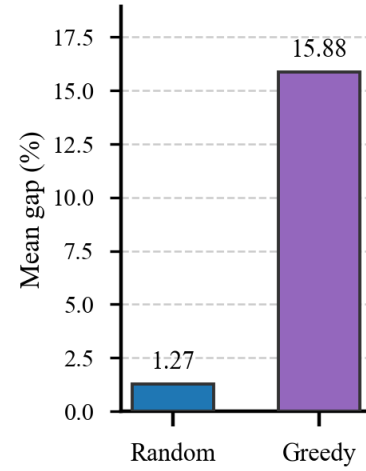
Location: Section V-B, Stage-I initialization-sensitivity analysis

To examine the effect of the initial deployment set in Algorithm 1, we compare Random, Density, DistSum, marginal Greedy, and Diverse initialization over 50 runs. For each configuration, the reported gap is the percentage difference between the final cost and the best final cost observed under the same server/user setting. As shown in Fig. 9, the five strategies reach the same best final cost in nearly all fixed-130-user cases; the only nonzero entry is the 2.10% mean gap of Random at 10 servers. Under the 10-server/150-user setting, Random obtains a 1.27% mean gap, whereas marginal Greedy reaches 15.88%. These results show that CLS is generally stable across initializations and that random initialization avoids a systematic preference for a poorer local optimum.

Fig. 9. Initialization sensitivity of CLS under two server/user settings.



(a) 130 users.



(b) 10 servers/150 users.

Response Fig. R2-1. CLS initialization sensitivity across the fixed-130-user configurations and the 10-server/150-user stress case.

Mean gap to the best final CLS cost over 50 runs

Configuration	Random mean gap	Marginal-greedy gap	Other initializers
5/130	0.00%	0.00%	0.00%
10/130	2.10%	0.00%	0.00%
15/130	0.00%	0.00%	0.00%
20/130	0.00%	0.00%	0.00%
10/150	1.27%	15.88%	Not used in the focused pairwise panel

Comment 2: Reliability-related QoS metrics**Reviewer Comment**

In Eq. (6), the QoS constraint only constrains the end-to-end latency upper bound D_i . However, packet loss, reliability, and service interruption probability are not considered. Since MEC systems for latency-sensitive applications usually require reliability guarantees, the manuscript is suggested to discuss the impact of ignoring reliability-related QoS metrics.

Response

Equation (6) retains end-to-end latency as the primary QoS requirement because the present optimization focuses on the cost-latency trade-off. The accompanying model discussion now clarifies that, for reliability-critical applications, latency compliance can be complemented by packet-loss, link-availability, and service-interruption constraints so that low-latency solutions are not selected when continuity is insufficient. The Conclusion also identifies reliability-aware QoS constraints as a direct extension of the framework.

Revised Manuscript Text*Location: Section III-C and Conclusion*

Equation (6) provides latency-oriented QoS by requiring the end-to-end latency of a served request to remain below the user's tolerable bound $\tilde{D}_l$. For reliability-critical applications, latency compliance can be complemented by packet-loss, link-availability, and service-interruption constraints, preventing a low-latency solution from being selected when service continuity is insufficient.

Future work will extend the framework with reliability-aware QoS constraints for packet loss, link availability, and service interruption, together with learning-based demand prediction and multi-domain coordination.

Comment 3: PSP/NSGA-II hyperparameters**Reviewer Comment**

The proposed PSP algorithm relies on NSGA-II for multi-objective optimization. Nevertheless, the manuscript lacks a detailed explanation of several key hyperparameters, such as population size N , mutation probability, crossover probability, and maximum generation number G .

Response

A complete parameter table has been added. The experiments use $N = 50$, $G = 200$, simulated binary crossover with $p_c = 0.9, \eta_c = 15$, polynomial mutation with $p_m = \frac{1}{k|\mathcal{S}|}, \eta_m = 20$, hybrid-score weights $\alpha = \beta = 0.5$, per-server service capacity $V_j = 4$, and anchor ratio $\rho = 0.5$. The revised algorithm text also states the non-dominated rank and crowding-distance selection rule and the explicit capacity-repair procedure.

Revised Manuscript Text*Location: Section V-A and Table II*

The evolutionary settings used for PSP are summarized in Table II. The deterministic anchor size is defined proportionally as $\varpi_j = \lceil \rho V_j \rceil, \rho = 0.5$. Hence, for the experimental capacity $V_j = 4$, two slots retain the highest-scoring services and two slots preserve stochastic exploration. This equal allocation provides an interpretable exploitation-exploration balance and applies directly to servers with different capacities.

PSP/NSGA-II parameter settings

Parameter	Setting	Role
Population size	$N = 50$	Number of individuals
Maximum generations	$G = 200$	Evolutionary iterations
SBX crossover	$p_c = 0.9, \eta_c = 15$	Crossover probability and index
Polynomial mutation	$p_m = \frac{1}{k \mathcal{S} }, \eta_m = 20$	Mutation probability and index
Hybrid-score weights	$\alpha = \beta = 0.5$	Cost-demand balance
Service capacity	$V_j = 4$	Services per server
Anchor ratio	$\rho = 0.5$	Deterministic capacity share

Comment 4: Definition and normalization of Q

Reviewer Comment

In Fig. 5 and Fig. 7, the performance metric Q (normalized) is presented, but its exact normalization process and mathematical definition are not sufficiently described.

Response

The revised manuscript defines min-max normalization over the pooled solutions of all compared methods within each server/user configuration using $\hat{C} = \frac{C - C_{\min}}{C_{\max} - C_{\min}}$ and $\hat{D} = \frac{D - D_{\min}}{D_{\max} - D_{\min}}$. The scalar evaluation score is $Q = \lambda \hat{C} + (1 - \lambda) \hat{D}, \lambda = 0.5$. Lower Q denotes a better balanced cost-delay solution. The text further clarifies that Q is applied only after multi-objective optimization for scalar comparison and is not an objective used to generate the Pareto population.

Revised Manuscript Text

Location: Section V-A, normalization and scalar evaluation

For each server/user configuration, cost and delay are normalized over the pooled solutions of the compared methods as

$$\hat{C} = \frac{C - C_{\min}}{C_{\max} - C_{\min}}$$

$$\hat{D} = \frac{D - D_{\min}}{D_{\max} - D_{\min}}$$

and the scalar evaluation score is

$$Q = \lambda \hat{C} + (1 - \lambda) \hat{D}, \lambda = 0.5$$

Lower Q indicates a better balanced solution. Hypervolume (HV) measures the dominated objective-space volume relative to the reference point (1.1,1.1) and is maximized, whereas inverted generational distance (IGD) measures the mean distance from the common non-dominated reference front to a method's front and is minimized.

Comment 5: Rationale for selecting the deterministic anchor size

Reviewer Comment

The proposed hybrid initialization mechanism in Algorithm 2 introduces the parameter ϖ_j to control the deterministic anchor size. However, the rationale behind selecting ϖ_j is unclear.

Response

The deterministic anchor is now defined proportionally as $\varpi_j = [\rho V_j], \rho = 0.5$. This rule assigns half of each server's capacity to the highest-ranked deterministic services and reserves the remaining half for stochastic selection from lower-ranked candidates. It therefore provides an interpretable exploitation-exploration balance and scales automatically with heterogeneous server capacities. In the reported experiments, $V_j = 4$ gives $\varpi_j = 2$; a capacity of 8 gives $\varpi_j = 4$.

Revised Manuscript Text*Location: Section IV-B and Section V-A*

PSP uses a hybrid initialization before the NSGA-II evolutionary loop. For each server, it constructs one ranking that favors low provisioning cost and another that favors services requested frequently by locally assigned users. The two rankings are fused by a hybrid score. High-scoring services form a deterministic anchor set, and the remaining capacity is filled by stochastic sampling from lower-ranked candidates. The resulting base configuration combines exploitation of high-quality candidates with exploration of alternative service combinations.

Let $\alpha, \beta \in [0,1]$, $\alpha + \beta = 1$ balance provisioning cost and local demand. Let ϖ_j ($0 < \varpi_j \leq V_j$) denote the deterministic anchor size on edge server m_j . The hybrid score of candidate service s_w is

$$H_j(s_w) = \alpha w_{\text{cost},j}(s_w) + \beta w_{\text{req},j}(s_w)$$

where $w_{\text{cost},j}$ and $w_{\text{req},j}$ are descending rank scores derived from provisioning cost and the request frequency of users assigned to m_j , respectively. We set $\varpi_j = \lceil \rho V_j \rceil$, $\rho = 0.5$, reserving half of the capacity for high-ranked deterministic services and half for stochastic selection from the remaining candidates.



Response Fig. R2-2. Hybrid initialization showing deterministic anchoring and stochastic filling.

Comment 6: Generalization to different geographical distributions and larger MEC settings

Reviewer Comment

In Section V, all experiments are conducted using a dataset collected within approximately a 9 km region around Xizhimen Subway Station in Beijing. However, the manuscript does not discuss the generalization capability of the proposed framework under different geographical distributions, heterogeneous traffic densities, or larger-scale MEC environments.

Response

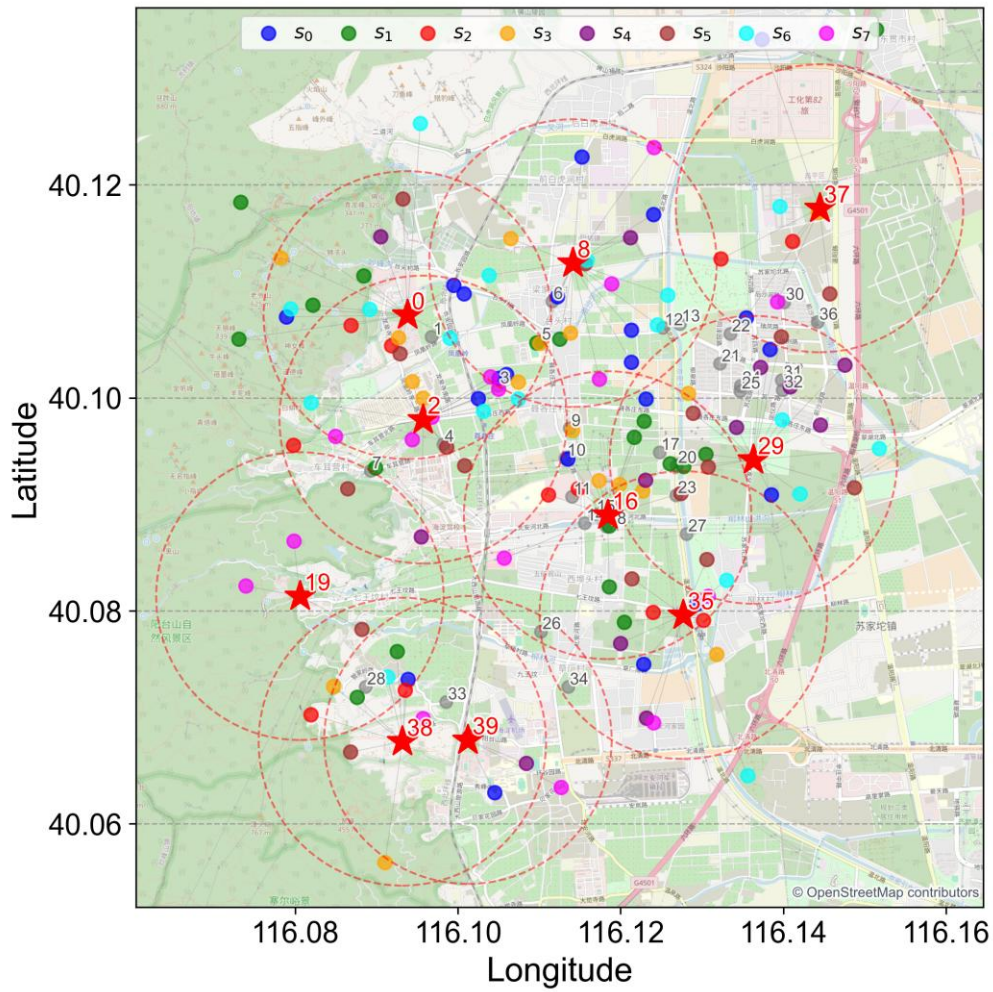
We conducted an additional complete two-stage experiment in a different real Beijing region using a pool of 2,215 deduplicated base-station coordinates. The evaluated instance contains 40 candidate base stations, 10 deployed servers, 130 users, and 8 service types. It changes the real base-station topology, doubles the candidate set relative to the primary experiment, and exhibits a more heterogeneous coverage-density structure.

In Stage I, CLS obtained a cost of 2,304.7670, compared with 6,150.5741 for the best non-CLS initialization-only result, a 62.53% reduction. In Stage II, the three-seed means for PSP were $HV = 1.0116$, $IGD = 0.0129$, and Best $Q = 0.2678$. PSP achieved the best HV and IGD in each seed and the best mean values of all three

metrics. The DQN baseline obtained mean Best $Q = 0.5517$, whereas PSP obtained 0.2678, corresponding to a 51.45% reduction. These results demonstrate that the complete framework remains effective under a different real deployment topology, a larger candidate set, and heterogeneous spatial demand.

Response-only real-region experiment configuration

Property	Value
Real Beijing station pool	2,215 deduplicated coordinates
Candidate stations / deployed servers	40 / 10
Users / service types	130 / 8
Coverage-density coefficient of variation	0.3359



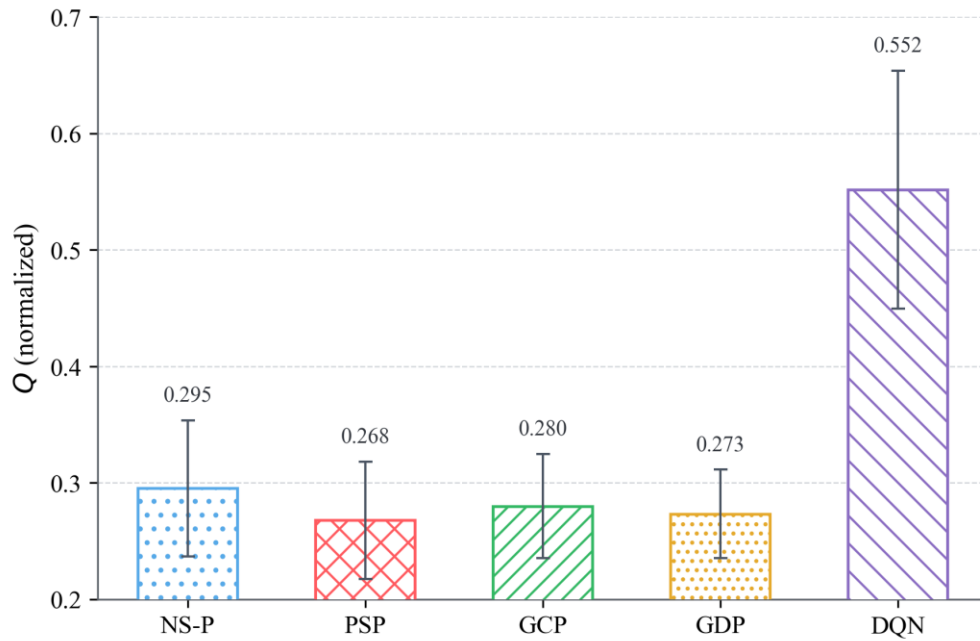
Response Fig. R2-3. Real-region topology, heterogeneous user requests, and CLS-selected edge-server locations.

Stage-I results in the new real-region instance

Stage-I method	Objective (lower is better)
CLS	2,304.7670
Best random trial	6,150.5741
Random-trial mean	13,540.5175
Density	22,206.4751
Distance-Sum	28,513.3397
Marginal Greedy	28,513.3397
Density-Diverse	19,031.7959

Stage-II results over seeds 42, 43, and 44

Method	HV mean $\pm$ std	IGD mean $\pm$ std	Best Q mean $\pm$ std
NS-P	0.9574 $\pm$ 0.0840	0.0498 $\pm$ 0.0328	0.2953 $\pm$ 0.0584
GCP	0.9799 $\pm$ 0.0691	0.0446 $\pm$ 0.0158	0.2800 $\pm$ 0.0447
GDP	0.9742 $\pm$ 0.0522	0.0408 $\pm$ 0.0234	0.2733 $\pm$ 0.0380
PSP	1.0116 $\pm$ 0.0579	0.0129 $\pm$ 0.0055	0.2678 $\pm$ 0.0503



Response Fig. R2-4. Mean Best Q in the real-region experiment; bars report the three-seed means.

Balanced-solution comparison including DQN

Method	Best Q mean	Std.
NS-P	0.2953	0.0584
GCP	0.2800	0.0447
GDP	0.2733	0.0380
PSP	0.2678	0.0503
DQN	0.5517	0.1021

Comment 7: Learning-based service-placement baseline**Reviewer Comment**

The comparison baselines in Stage II mainly include heuristic initialization strategies (GCP, GDP) and standard NSGA-II initialization. However, the manuscript does not compare against recent learning-based service placement methods, such as deep reinforcement learning or graph neural network based approaches.

Response

We included a Deep Q-Network provisioning baseline. Stage-II provisioning is represented as a sequence of server-slot decisions in which the Q-network selects a service type or an empty action from demand, deployment-cost, current-selection, and cost-delay preference information. The resulting service matrix is evaluated by exactly the same cost and delay functions as the other methods. The implementation uses one hidden layer with 64 units and trains for 320 episodes for each preference weight in $\{0.1, 0.3, 0.5, 0.7, 0.9\}$; the learning rate is 7×10^{-4} , the discount factor is 0.98, the mini-batch size is 64, and the replay capacity is 12,000.

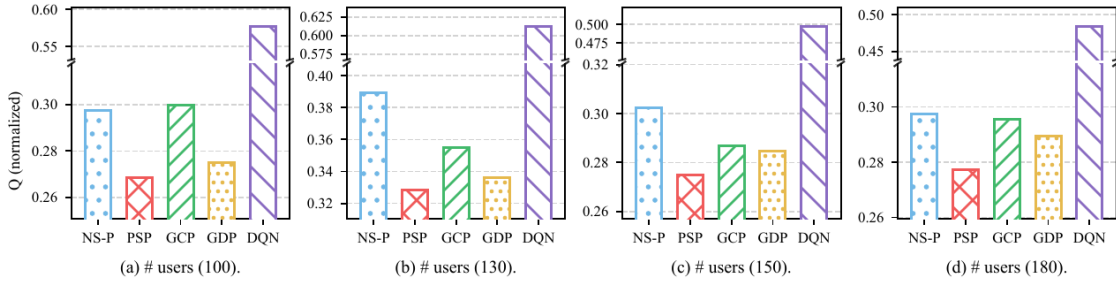
Across both primary Stage-II experiment series, PSP attained the lowest normalized Q in every reported case. In the representative 10-server/130-user setting, PSP achieved $Q = 0.3282$, compared with $Q = 0.6125$ for DQN. Because DQN returns preference-conditioned deployment points rather than an evolutionary population, it is compared through the common scalar Q measure and is not presented as a Pareto curve.

Revised Manuscript Text

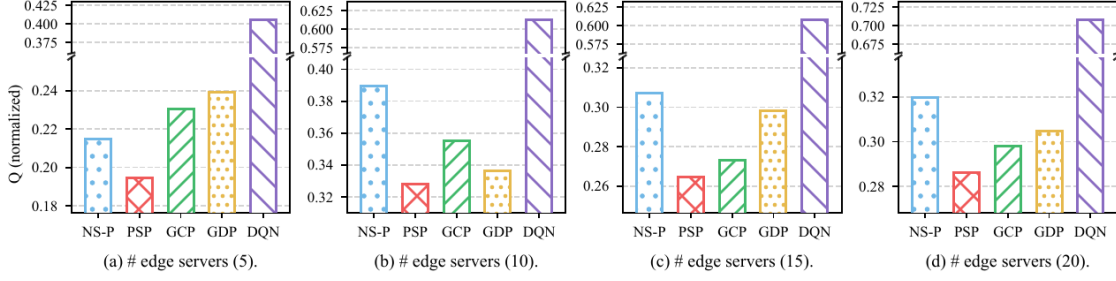
Location: Section V-A and Section V-B

Deep Q-Network Provision (DQN): A Q-network sequentially selects a service or an empty action for each server slot using demand, deployment cost, and a cost-delay preference. The resulting deployment is evaluated by the same objectives as the other methods.

We evaluate Stage II under two complementary configurations: (i) 10 deployed servers with 100, 130, 150, and 180 users, and (ii) 130 users with 5, 10, 15, and 20 deployed servers. Figures 5 and 7 report the minimum normalized score Q obtained by each method. PSP achieves the lowest Q in every reported case across both experimental series. Its hybrid initialization consistently improves upon random NS-P and the single-criterion GCP and GDP initializations; PSP also achieves lower Q than DQN across all tested scales.



Response Fig. R2-5. Stage-II Best Q with 10 deployed servers and increasing user populations.



Response Fig. R2-6. Stage-II Best Q with 130 users and increasing deployed-server counts.

Direct PSP-DQN comparison using the common normalized Q

Series	Servers/users	PSP Q	DQN Q	PSP reduction
Fixed 10 servers	10/100	0.2686	0.5770	53.5%
Fixed 10 servers	10/130	0.3282	0.6125	46.4%
Fixed 10 servers	10/150	0.2749	0.4972	44.7%
Fixed 10 servers	10/180	0.2774	0.4840	42.7%
Fixed 130 users	5/130	0.1945	0.4052	52.0%
Fixed 130 users	10/130	0.3282	0.6125	46.4%
Fixed 130 users	15/130	0.2646	0.6077	56.5%
Fixed 130 users	20/130	0.2864	0.7077	59.5%

Comment 8: Quantitative Pareto metrics

Reviewer Comment

In Fig. 8, the Pareto fronts of different algorithms are illustrated, but no quantitative Pareto evaluation metrics (e.g., Hypervolume, IGD, Spacing, or Spread) are provided. Relying only on visual comparison may not be sufficiently rigorous.

Response

We added Hypervolume (HV) and Inverted Generational Distance (IGD), together with Best Q , for the representative 10-server/130-user configuration. HV measures dominated objective-space volume relative to the common reference point (1.1,1.1) and is maximized; IGD measures the mean distance from the common

non-dominated reference front to a method's front and is minimized. PSP achieved $HV = 0.9470$, $IGD = 0.0016$, and Best $Q = 0.3282$, outperforming NS-P, GCP, and GDP on all three measures.

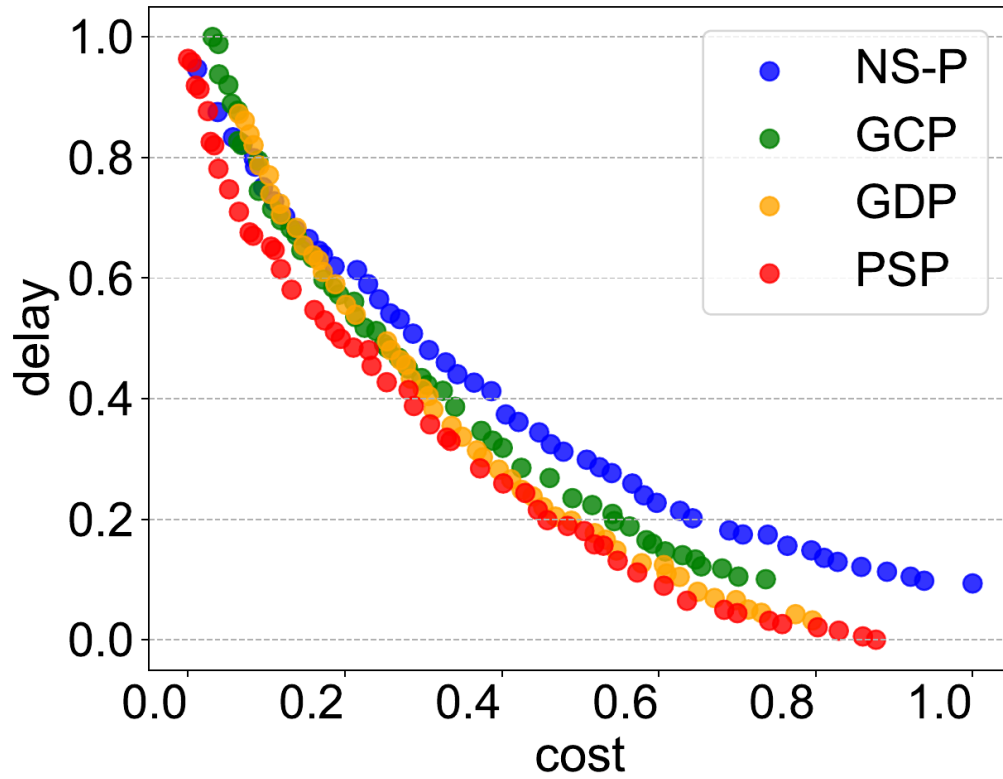
DQN provides a limited set of preference-conditioned solutions rather than an equal-cardinality population-based Pareto front. It is therefore evaluated through the common Best Q measure, while HV and IGD are reported for the four population-based strategies.

Revised Manuscript Text

Location: Section V-A and Section V-B

Lower Q indicates a better balanced solution. Hypervolume (HV) measures the dominated objective-space volume relative to the reference point $(1.1,1.1)$ and is maximized, whereas inverted generational distance (IGD) measures the mean distance from the common non-dominated reference front to a method's front and is minimized.

Figure 8 compares the cost-delay fronts generated by the four NSGA-II-based provisioning strategies with 130 users. Table III gives the numerical results for the 10-server setting. PSP attains the largest HV (0.9470), the smallest IGD (0.0016), and the smallest Q (0.3282), confirming both broader objective-space coverage and closer convergence to the common reference front. DQN produces a limited set of scalarized solutions rather than a population-based Pareto front; it is therefore compared through Q and is not included in the HV/IGD comparison.



Response Fig. R2-7. Pareto fronts for the representative 10-server/130-user configuration.

Quantitative Pareto and balanced-solution metrics

Method	HV (higher)	IGD (lower)	Best Q (lower)
NS-P	0.8191	0.0785	0.3894
GCP	0.8596	0.0492	0.3550
GDP	0.8945	0.0326	0.3363
PSP	0.9470	0.0016	0.3282
DQN	N/A	N/A	0.6125

Comment 9: Suggested references**Reviewer Comment**

The following works are closely related to MEC and server placement, and thus should not be overlooked. [1] Latency-Aware Service Deployment and Peer Offloading: A Long-Term Optimization Framework for Satellite Edge Computing. [2] Latency-Aware Task Offloading in Multi-Tier SAGIN With FSO-Enabled Mobile Edge Computing. [3] Novel Breakout Local Search for Offloading Tasks in Multi-Tiered Cloud Environment Considering Transmission and Processing. [4] Long-Term Max-Min Fairness Guarantee Mechanism for Integrated Multi-RAT and MEC Networks. [5] Dynamic Energy Cost Conservation for Distributed Edge Clouds Utilizing Online Mini-Batch Learning. [6] Mobile Edge Computing Offloading for Static Users in a Free Space Optical Communications-Enabled Satellite-Air-Ground Integrated Network.

Response

Following the Associate Editor's guidance, we evaluated the six works individually and incorporated four studies with direct methodological relevance: Feng et al. on latency-aware service deployment and peer offloading; Kato et al. on breakout local search for transmission- and processing-aware offloading; Jing et al. on long-term max-min fairness in integrated multi-RAT/MEC networks; and Jing et al. on online mini-batch learning for dynamic energy-cost conservation in distributed edge clouds. The Related Work section now explains how these studies complement the present focus on interpretable server planning and capacity-constrained multi-objective service provisioning.

The two remaining studies specifically address task offloading in FSO-enabled satellite-air-ground integrated networks. Their network architecture, optical-link assumptions, and decision variables differ substantially from the terrestrial MEC service-provisioning model considered here; consequently, they were not included solely to increase the citation count.

Revised Manuscript Text

Location: Section II-C, Related Work

Optimization and learning methods also address complementary MEC resource-management objectives. Feng et al. [33] jointly optimized latency-aware service deployment and peer offloading over multiple timescales. Kato et al. [34] applied breakout local search to transmission- and processing-aware task offloading in a multi-tier cloud environment. Jing et al. [35] studied long-term max-min fairness for task splitting and resource allocation in integrated multi-RAT/MEC networks, while Jing et al. [36] used online mini-batch learning for dynamic energy-cost conservation in distributed edge clouds. These studies address dynamic offloading, fairness, or energy management; the present work focuses on interpretable server planning and capacity-constrained multi-objective service provisioning under a common cost-latency model.

1. C. Feng, M. Yang, Z. Jing, T. Q. S. Quek, and M. Mei, "Latency-Aware Service Deployment and Peer Offloading: A Long-Term Optimization Framework for Satellite Edge Computing," *IEEE Internet of Things Journal*, vol. 13, no. 1, pp. 405-417, 2026.
2. M. Kato, T. K. Rodrigues, and S. Verma, "Novel Breakout Local Search for Offloading Tasks in Multi-Tiered Cloud Environment Considering Transmission and Processing," *Proc. IEEE IC-NIDC*, pp. 269-274, 2025.
3. Z. Jing, Q. Yang, M. Qin, J. Li, and K. S. Kwak, "Long-Term Max-Min Fairness Guarantee Mechanism for Integrated Multi-RAT and MEC Networks," *IEEE Transactions on Vehicular Technology*, vol. 70, no. 3, pp. 2478-2492, 2021.
4. Z. Jing, X. Wang, Q. Yang, M. Mei, and Y. Wu, "Dynamic Energy Cost Conservation for Distributed Edge Clouds Utilizing Online Mini-Batch Learning," *Proc. IEEE PIMRC*, pp. 1-6, 2023.

Closing

We appreciate the Editor's and Reviewers' comments, which have improved the clarity, reproducibility, and experimental support of the manuscript. We hope that the revised manuscript is now suitable for further consideration.

Sincerely,

The Authors